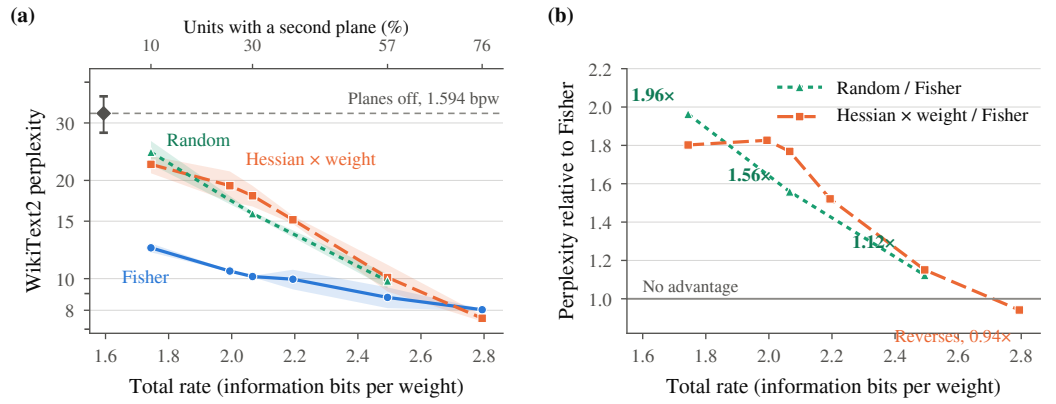
(a)
Units with a second plane (%)
10
30
57
76
Planes off, 1.594 bpw
Random
Hessian × weight
Fisher
WikiText2 perplexity
30
20
15
10
8
1.6
1.8
2.0
2.2
2.4
2.6
2.8
Total rate (information bits per weight)
(b)
Random / Fisher
Hessian × weight / Fisher
1.96×
1.56×
1.12×
No advantage
Reverses, 0.94×
Perplexity relative to Fisher
2.2
2.0
1.8
1.6
1.4
1.2
1.0
1.6
1.8
2.0
2.2
2.4
2.6
2.8
Total rate (information bits per weight)